

Built-in R Features - Data Structures

R contains quite a few useful built-in functions to work with data structures. Here are some of the key functions to know:

- `seq()`: Create sequences
- `sort()`: Sort a vector
- `rev()`: Reverse elements in object
- `str()`: Show the structure of an object
- `append()`: Merge objects together (works on vectors and lists)

In [8]:

```
# seq(start,end,step size)
seq(0, 100, by = 3)
```

Out[8]:

```
 0  3  6  9 12 15 18 21 24 27 30 33 36 39 42 45 48 51 54 57 60 63 66 69 72 75
78 81 84 87 90 93 96 99
```

In [10]:

```
v <- c(1,4,6,7,2,13,2)
```

In [14]:

```
sort(v)
```

Out[14]:

```
1 2 2 4 6 7 13
```

In [15]:

```
sort(v,decreasing = TRUE)
```

Out[15]:

```
13 7 6 4 2 2 1
```

In [18]:

```
v2 <- c(1,2,3,4,5)
rev(v2)
```

Out[18]:

```
5 4 3 2 1
```

In [20]:

```
str(v)
```

```
num [1:7] 1 4 6 7 2 13 2
```

In [22]:

```
append(v,v2)
```

Out[22]:

```
1 4 6 7 2 13 2 1 2 3 4 5
```

In [23]:

```
sort(append(v,v2))
```

Out[23]:

```
1 1 2 2 2 3 4 4 5 6 7 13
```

Data Types

- `is.*()`: Check the class of an R object
- `as.*()`: Convert R objects

In [1]:

```
v <- c(1,2,3)  
is.vector(v)
```

Out[1]:

```
TRUE
```

In [2]:

```
is.list(v)
```

Out[2]:

```
FALSE
```

In [3]:

```
as.list(v)
```

Out[3]:

```
1. 1  
2. 2  
3. 3
```

In [4]:

```
as.matrix(v)
```

Out[4]:

1
2
3

Great! That's if for the built-in data structure functions! These will com ein handy so try to remember them for the exercises!

